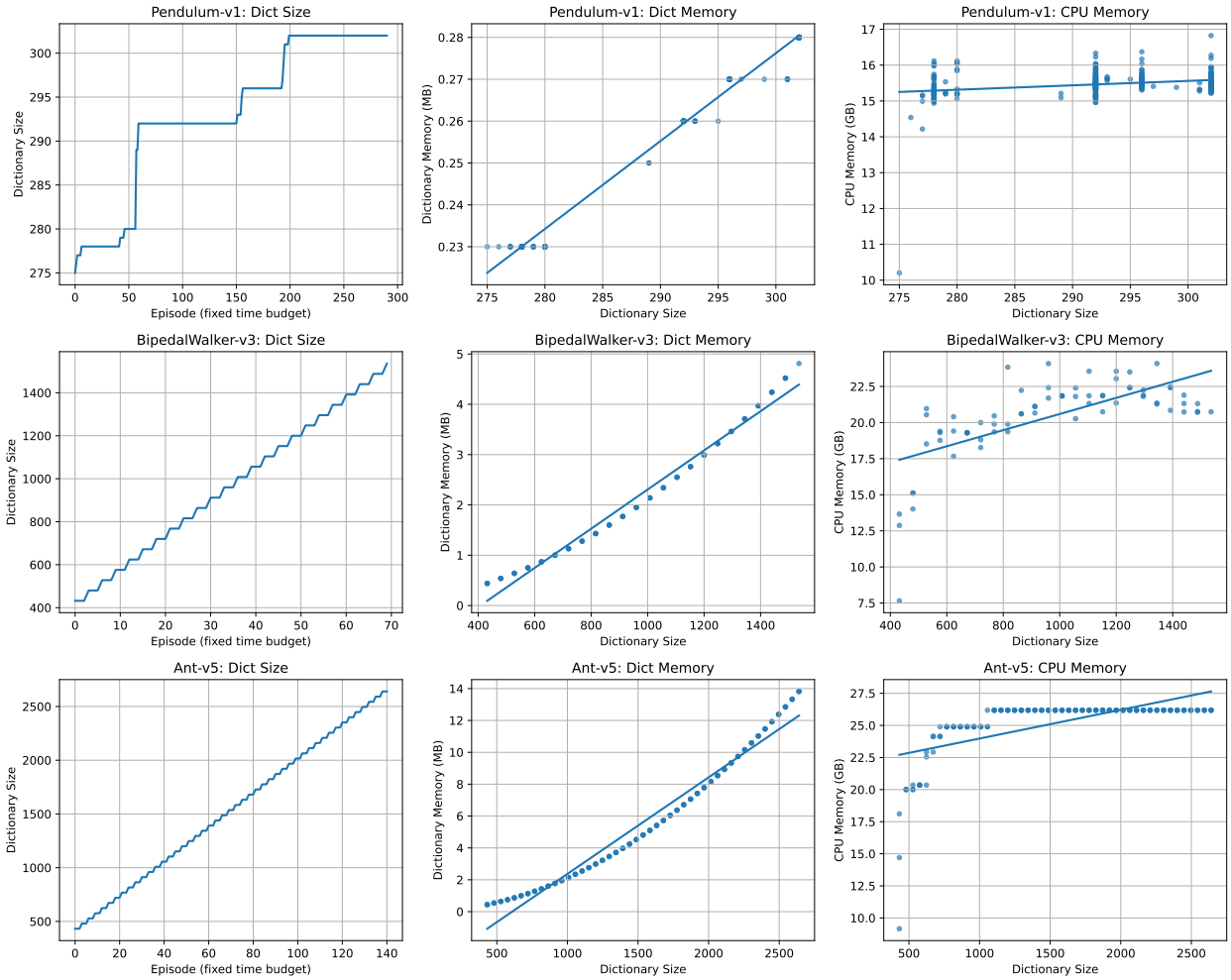
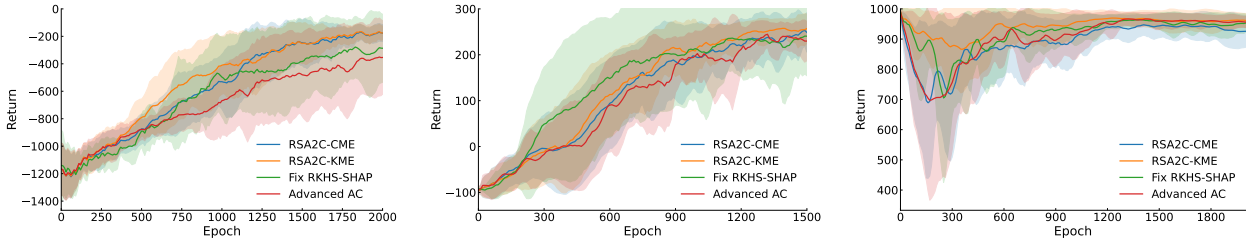


Figure 1: The performance of computational cost.



(a) `Pendulum-v1` (b) `BipedalWalker-v3` (c) `Ant-v5`

Figure 2: The performance of different type of RKHS-SHAP.

Table 1: Performance of three tasks on high noise.

		0.05	0.1	0.5
Pendulum-v1	RSA2C-CME	-140.50 ± 57.29	-463.32 ± 156.59	-841.46 ± 81.99
	RSA2C-KME	-255.13 ± 198.06	-354.09 ± 50.91	-858.21 ± 205.39
BipedalWalker-v3	RSA2C-CME	246.66 ± 25.66	232.34 ± 5.70	-12.33 ± 29.02
	RSA2C-KME	241.83 ± 10.07	185.04 ± 15.86	-23.46 ± 29.91
Ant-v5	RSA2C-CME	970.91 ± 6.43	965.67 ± 12.33	-830.79 ± 389.33
	RSA2C-KME	956.38 ± 5.33	967.76 ± 3.78	-1109.16 ± 140.58

Table 2: Performance of ablation study with noise.

		0.001	0.005	0.01
Pendulum-v1	Advanced AC	-420.87 ± 241.46	-355.55 ± 223.12	-165.73 ± 50.81
	RKHS-AC	-267.98 ± 181.49	-278.38 ± 35.93	-310.91 ± 185.06
BipedalWalker-v3	Advanced AC	271.57 ± 9.78	264.98 ± 22.18	255.36 ± 3.60
	RKHS-AC	242.18 ± 48.48	193.85 ± 95.23	176.95 ± 118.93
Ant-v5	Advanced AC	956.60 ± 34.01	952.28 ± 9.36	953.54 ± 25.82
	RKHS-AC	995.60 ± 4.61	996.37 ± 4.94	995.70 ± 4.31

Table 3: Time-to-solution analysis across environments. Convergence epochs are estimated from learning curves.

Environment	**Algorithm**	**Epochs**	**Runtime (ms)**	**Time-to-Solution (s)**
Pendulum-v1	SAC (GPU)	80	7346.2	**587.7**
	RSA2C-CME	1100	685.6	754.1
	RSA2C-KME	1400	667.7	934.7
	PPO (GPU)	1500	2713.3	4069.9
BipedalWalker-v3	RSA2C-KME	1200	1354.7	**1625.7**
	RSA2C-CME	1200	1448.3	1737.9
	PPO (GPU)	500	8119.8	4059.9
	SAC (GPU)	450	23982.8	10792.2
Ant-v5	RSA2C-KME	500	1358.7	**679.4**
	RSA2C-CME	500	1700.4	850.2
	SAC (GPU)	1400	9623.5	13472.9
	PPO (GPU)	–	1183.0	–